

Open VSP Analysis & Integration into MATLAB Analysis Code Procedures

Aero Analysis (subsonic and supersonic if applicable)

1. Open OpenVSP and load in the configuration you want to analyze.
 - The aspect ratio of each facet should be near square (by visual inspection). This correlates to a W/U ratio of between 41/21 to 61/41 depending on wing size. (61/41 tends to be for larger horizontal surfaces)
 - Locations
 - Num W: under Section tab
 - Num U: under Section tab
 - Tip cluster: under Section tab
 - Tip = 0.25
 - Root = 1
2. Analysis --> Aero --> VSPAERO
 - Overview
 - Geometry: shown
 - Case Setup: Vortex Lattice
 - Reference Area Lengths: From Model
 - Select the wing you are using – openVSP just selects the first wing it finds which may not be your wing. The numbers tell you which element in Geom Browser
 - Ie: 3_Wing_Geom could be the first wing but the 3rd geometry item
 - Alpha Num pts 18
 - Alpha start -5
 - Alpha end 12
 - Mach start (single value as you may run multiple times with different Mach's)
 - If subsonic the M0 from the xlsx sheet
 - If supersonic something like 0.3
 - Mach Num pts 1
 - ReCref Start: the values calculated by the trade study's spreadsheet (low speed ~1e7, Mach >0.8 ~1e8)
 - The main focus is to have the correct magnitude
3. Run Launch Solver
 - Should see results manager and a plot of the drag polar
 - Do not click “export to .csv” button – just running the analysis will output the files you need

4. Go into the directory where the results output (usually the same folder as the vsp file is) and open the .polar file. If this does not work (it doesn't open with excel you may have to right click and "open with" and select excel. You should be able to set that all .polar files to "open with excel" for the future). If that doesn't work, open excel app and go to file->open->browse (select all files by the file name box) and click on the polar file
 - When opening the .polar file it may walk you through how to import text to excel format. If it does not: highlight the whole first column --> Data --> Text to Columns --> finish
 - This should correctly format the Excel sheet
 - Resave this file as an Excel file (.xlsx or .xlsm) and be sure to resave and rename this file with the specific configuration & Mach. Otherwise each time you run the analysis it will overwrite the old data
 - Ex: c172_sw40_M03_aero.xlsm (cessna 172 with a sweep of 40 and at Mach 0.3 aero analysis)

Parasite Analysis (subsonic & supersonic (supersonic will need to run at 0.3, 1.05, and M_max)

1. Open OpenVSP and load in the configuration you want to analyze. Show only the external surfaces
2. Analysis --> Aero --> Parasite
 - Overview
 - Geometry: Shown
 - The units should be in "ft"
 - Laminar Cf Eqn: Blasius:
 - Turbulent Cf Eqn: Power Blasius
 - Reference Area: From model
 - Alt = the altitude you are expecting to fly at for the mission segment corresponding with that mach
 - $V_{inf} = M \cdot a$
 - The Mach is whatever much you want to do analysis for – supersonic make sure you know which Mach this run is
 - Percent laminar = 0
 - Make sure each of your Q values match the values from 9 - ASEN 4138 Using Open VSP for Aero Analysis.pdf (you may have to scroll over a little on openvsp window to find the Q input box)
 - Ref wing – the wing you want to analyze (check to make sure the correct wing is selected)
 - REMINDER: THE ENGINES MUST BE SOLID BODIES
 - Run: Calculate CD0

- It has run when boxes in the bottom right have values or have changed values
- Download as csv file --> name it based on which configuration you are testing (it will not overwrite if you name them differently) I recommend you name them with the specific configuration & Mach
 - Ex: c172_sw40_M03_parasite.csv (cessna 172 with a sweep of 40 and at Mach 0.3 parasite analysis)

Wave Drag (supersonic only (will need to do it for M_max and 1.05))

1. Open OpenVSP and load in the configuration you want to analyze.
2. Analysis --> Aero --> Wave
 - Setup
 - Geometry: Shown
 - Reference Area: From Model
 - Ref Wing: the wing you want to analyze
 - Mach
 - The Mach is whatever Mach you are choosing – supersonic make sure you know which Mach this run is
 - Go into File -> “...” and change the name of the file output
 - I recommend you name them with the specific configuration & Mach
 - Ex:T38_sw40_M125_wave.txt (T38 with a sweep of 40 and at Mach 1.25 wave analysis)
 - REMINDER: THE ENGINES MUST BE FLOWTHROUGH BODIES
3. Run parasite analysis: Start
 - Cd0_w box should be populated after running
4. Go into the directory where it the results output (usually the same folder as the vsp file is) and check the .txt file is present
5. Reload the open vsp file before running any other analysis even if it is wave again or there are issues with the following analysis results

Using in Mission Performance Analysis

1. Download the openVSP functions folder & extract files into the directory you are running your code from. None of these should overwrite any previous functions
2. Fill in the information in the “%% ALL MANUAL INPUTS LIVE HERE” section which negates the need to continually update your Design Input.xlsx file.
 - a. The code allows typing an array of different values or file names so that you only have to input everything once and you can choose the iteration with the variable “Config which is the Configuration number.

- i. Caution: make sure all the information for a configuration matches up so you don't by accident run the Sref of config1 with the analysis files of config 3
 - ii. Also, if you are subsonic and inputting multiple configurations make sure to type {"", "" ...} for the supersonic file names or else you will get an error
- 3. Copy all mission segments from your previous Mission Performance file into here
 - a. For each run you will need to keep track of the names and location of all your openVSP outputs. I would recommend naming them according to the Mach number, the analysis and the configuration. Ex: "T38_sw40_M125_wave.txt" (T38 with a sweep of 40 and at Mach 1.25 wave analysis)
 - b. For subsonic there are 2 files per configuration you need to input into the Mission Performance: a (.xlsx) file from aero analysis and a (.csv) file from parasite analysis.
 - c. For supersonic there are 6: the aero(.xlsx) and parasite (.csv) from subsonic run at 0.3 Mach, a parasite file (.csv) and wave drag file (.txt) from $M = 1.2$, and a parasite (.csv) and wave drag file (.txt) from you design mach specified in your Main Inputs tab of your input file.
 - d. Subsonic leave the last 4 names blank.
 - e. If you have your files in a different location than where the Mission Performance is running from you will need to include the path in the file name